PRODUCTION AI CASE STUDY

ClinicDocs AI

FLAGSHIP OPEN-SOURCE CONVERSATIONAL RAG

Chat with Clinical Policies with Zero Hallucinations & Exact Page Citations

A production-grade, privacy-first Conversational RAG web application built to turn static clinic policies, operating hours, and patient guidelines into grounded answers with 1-indexed citations and two-layer refusal guardrails.

clinicdocs-ai.streamlit.app — Live Session Active

10/10 EVALS PASSED

User: What are the clinic operating hours on Friday?

ClinicDocs AI: Latency: 0.82s

On Friday, Noor Care Clinic operates in two shifts:
- **Morning Shift:** 9:00 AM – 1:00 PM
- **Evening Shift:** 3:00 PM – 8:00 PM

(Closed between 1:00 PM and 3:00 PM for Friday prayers).

Noor_Care_Clinic_FAQ.pdf — Page 1 Match: 78.2%

70%	< 25 ms	0 Tokens	100%
RAM Savings (ONNX)	Refusal Gate Speed	Wasted on OOD Queries	Grounded Accuracy

Architected & Built by Farwa Ali Nasir

Swipe right to inspect the engineering ➜

THE ENGINEERING PROBLEM

Why 90% of RAG Demos Fail in Production

Tutorial notebooks work with 5-line toy scripts, but real enterprise deployments hit severe memory, privacy, and hallucination constraints. Here is how we engineered around them:

✕ 1. Heavy PyTorch OOM Container Crashes

Standard `sentence-transformers` imports heavy PyTorch wheels (~800MB), instantly exceeding 1GB serverless RAM caps and crashing containers.

✓ FastEmbed ONNX (70% Less Memory)

Replaced PyTorch with pure quantized ONNX runtime (`bge-small-en-v1.5`). Cold-starts in 2.2s and peaks at just **423MB RAM** under full load.

✕ 2. Shared-Disk Data Contamination

Persistent Chroma/SQLite files stored on disk retain previous users' confidential medical files, causing multi-tenant data leakage.

✓ In-Memory Isolation with LRU Eviction

Vector collections live exclusively in process RAM (`chromadb.EphemeralClient`) tied to session UUIDs with an LRU cap (max 10) to prevent memory leaks.

✕ 3. Hallucinations & Wasted API Bills

Irrelevant queries trigger full LLM completions, wasting API budget while fabricating answers to off-topic or unanswerable medical questions.

✓ Two-Layer Refusal Gate (<25ms, 0 Tokens)

Cosine similarity threshold gate ($s < 0.48$) intercepts unrelated queries in 18ms with 0 LLM tokens, backed by strict XML prompt grounding.

SYSTEM ARCHITECTURE

The 4-Stage Production Pipeline

STAGE 01 — src/ingest.py

Ingestion & Normalization

- **1-Indexed Offsets:** Maps PyPDF `page_idx` to physical `page + 1` .
- **SHA-256 Digest:** Prevents redundant document re-indexing.
- **Scanned PDF Guard:** Density checks reject non-extractable PDFs.

STAGE 02 — FastEmbed ONNX

Quantized Embeddings

- **Model:** `BAAI/bge-small-en-v1.5` (384-dimensional).
- **Pure ONNX Runtime:** Strips 800MB PyTorch dependency.
- **Low Latency:** Cold-start in 2.22s, peaks at 136MB RAM delta.

STAGE 03 — src/retriever.py

Ephemeral Chroma Store

- **Zero Disk Retention:** Collections live in RAM (`EphemeralClient`).
- **Session Isolation:** Keyed to unique session UUIDs.
- **LRU Eviction:** Capped to 10 active collections to prevent bloat.

STAGE 04 — src/chain.py

Two-Layer Guardrails

- **Layer 1 Gate:** Cosine similarity threshold ($s \ge 0.48$).
- **Layer 2 Grounding:** XML `<document_context>` isolation.
- **Conditional Rewrite:** Rewrites history only on Turn 2+.

Designed for High Throughput & Multi-Tenant Safety

Swipe right to see real dialogue verification ➜

VERIFIED EVIDENCE

Grounded Answers vs 18ms Refusals

● Case 1: Grounded Multi-Turn Question with Physical Page Citation

Latency: 0.94s

User: How much is a specialist consultation, and what is the follow-up policy?

A consultation with a Specialist Doctor (Cardiologist, Pediatrician, Dermatologist) costs **PKR 3,000**. Follow-up visits within 7 days of the initial consultation receive a **50% discount (PKR 1,500)**.

Clinic_Fee_Schedule_2026.pdf — Page 2

Cosine Similarity: 81.4%

● Case 2: Layer 1 Refusal Gate (Semantically Distant Out-of-Domain Query)

Intercepted in 18ms (0 Tokens)

User: What is the capital city of France?

"I couldn't find this in the uploaded documents."

Max observed similarity: 0.361 < 0.48 calibrated threshold. Query halted immediately at vector retrieval stage. Zero LLM API cost incurred.

PRODUCTION CODE

The Core Engineering Implementation

src/retriever.py — Layer 1 Vector Refusal Gate — Python 3.11

```python
def retrieve_with_gate(
    vectorstore: Chroma, query: str, k: int = 4, threshold: float = 0.48
) -> tuple[list[Document], float, bool]:
    # 1. Compute cosine similarity across top-k chunks
    results = vectorstore.similarity_search_with_relevance_scores(query, k=k)
    max_sim = max([score for _, score in results]) if results else 0.0

    # 2. Intercept OOD queries at vector stage in <25ms (0 tokens billed)
    if max_sim < threshold:
        logger.info("Gate refused query '%s' (sim=%.3f < %.2f)", query, max_sim, threshold)
        return [], max_sim, False

    return [doc for doc, _ in results], max_sim, True
```

src/chain.py — XML Context Grounding & Injection Hardening — LangChain LCEL

```python
prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You are ClinicDocs AI. Answer ONLY using <document_context>.\\n"
     "If ungrounded, refuse with: 'I couldn't find this in the uploaded documents.'\\n"
     "Never fabricate medical treatments or unstated clinic policies.\\n\\n"
     "<document_context>\\n{context}\\n</document_context>"),
    MessagesPlaceholder(variable_name="chat_history"),
    ("human", "{question}")
])
```

100% Tested with Pytest • Full Source Available on GitHub

Swipe right for measured benchmarks & results ➜

EMPIRICAL TELEMETRY

Measured Benchmarks vs Baselines

10/10
Evaluation Suite
100% Grounded

< 25 ms
Refusal Speed
0 Tokens Used

423 MB
Peak Process RAM
70% Less Memory

2.22 s
Cold Start
82% Faster

ENGINEERING METRIC	CLINICDOCS AI	NAIVE PYTORCH RAG	PRODUCTION IMPACT
Embedding Runtime Memory	**136.2 MB (ONNX)**	850+ MB (PyTorch)	**70% Reduction**
Total Process RAM Footprint	**423.2 MB**	1,420 MB (OOM Risk)	**Fits 1GB Cap**
OOD Refusal Latency	**< 25 ms**	2,850 ms (Full LLM call)	**99.4% Faster**
Cost on Irrelevant Queries	**$0.00 (0 Tokens)**	Standard API Billing	**Zero Cost Waste**

FAN

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Live App & Repo Links in Comments 👇
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